

Roll No.

TEC-701

**B. TECH. (ECE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

PRINCIPLES OF MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Explain any *five* qualities of a manager in detail. (CO1)
(b) What are various goals or objectives of a management ? (CO1)
(c) Write any *three* definitions of management. (CO1)
2. (a) Evaluate the goals and objectives of planning using "SMARTER" tool. (CO2)
(b) Discuss major steps in planning in detail. (CO2)
(c) Discuss strength, weakness, opportunity and threat concepts in SWOT analysis. (CO2)

P. T. O.

(2)

3. (a) Define the following : (CO3)
- (i) Selection
 - (ii) Placement
- (b) Differentiate between centralization and decentralization of authority in an organization. (CO3)
- (c) Discuss 4 steps of organizing in detail. (CO3)
4. (a) Discuss any *five* principles of directing. (CO4)
- (b) What are various limitations in controlling ? (CO4)
- (c) What is controlling process ? Discuss 4 steps of controlling process. (CO4)
5. (a) What is recruitment ? Explain seven steps of recruitment process. (CO5, CO6)
- (b) Define Human Resource. Discuss challenges in HR Management. (CO5, CO6)
- (c) Mention any *two* definitions of human resource management. What is the importance of HRM ? (CO5, CO6)

Roll No.

TEC-731

**B. TECH. (ECE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) What is disaster awareness ? How does economic and social development increase disaster risk ?
(CO1 and CO5)
(b) Explain causes and effects of the following Natural Disasters :
(CO1 and CO5)
 - (i) Earthquake
 - (ii) Flood
 - (iii) Landslide
 - (iv) Tsunami
- (c) What kinds of disasters are prevalent in an urban area ? As a DM, what steps will you take to deal with them ?
(CO1 and CO5)
2. (a) Explain how Disaster preparedness is the first step in any disaster management process.
(CO3 and CO6)

P. T. O.

(2)

- (b) Differentiate between disaster management and disaster assessment.
What is Search and Rescue mission, in terms of disaster management ?
(CO3 and CO6)
- (c) Explain the role and functions of a disaster communication systems.
(CO3 and CO6)
- 3. (a) What are the operational applications of remote sensing for disaster management ? What about geological parameters like topography, geomorphology, soil lithology, etc ? Are there any studies on this subject ? Especially for landslides.
(CO3 and CO4)
- (b) Describe Vulnerability, Risk, Capacity and Early warning.
(CO3 and CO4)
- (c) What are the financial arrangements for disaster management ?
(CO3 and CO4)
- 4. (a) How is biodiversity loss associated with infrastructure development ?
How does environmental impact assessment help in reconciling the developmental needs ?
(CO2)
- (b) Disasters block the way to sustainable development. How ?
(CO2)
- (c) Define cyclones. What preventive measures can be taken at the time of cyclone ?
(CO2)
- 5. (a) What do you mean by structural and non-structural mitigation of Disaster Management ?
(CO5 and CO6)
- (b) What is the role of non-government and inter governmental agencies in disaster management ?
(CO5 and CO6)
- (c) Himalaya may be entering phase of irreversible decline because of its losses to its ecology. Discuss in the light of frequent disasters in the region.
(CO5 and CO6)

Roll No.

TEC-759

**B. TECH. (ECE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
INTERNET OF THINGS AND ITS APPLICATIONS**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) A sensor is connected to a microcontroller which has communication protocols to connect with the internet. Draw the possible communication through the internet of such a device. Can we use wireless sensor network topologies with this device ? (CO1)
- (b) Explain the constraints and challenges of the wireless sensor network. (CO1)
- (c) What are the functional blocks of IoT ? Explain with the help of a proper diagram. (CO1)
2. (a) Explain the outline structure of IoT Architecture. Take the example of agriculture in support of IoT architecture. (CO2)

P. T. O.

(2)

- (b) How IoT can be implemented for industrial infrastructure ? Explain your example in the context of the courier service industry. (CO2)
- (c) What is M2M ? What is the M2M value chain ? How can we connect M2M with IoT ? (CO2)
- 3. (a) Explain Deployment and operational view for IoT Architecture. (CO3)
- (b) Explain IoT Reference Model. (CO3)
- (c) Explain the operational view and information view in the context of IoT. (CO3)
- 4. (a) What is the role of IoT in Industry ? (CO4)
- (b) Explain the IoT system for Home Automation. (CO4)
- (c) How can IoT help in improving Health and Lifestyle ? (CO4)
- 5. (a) Explain Privacy and Trust in IoT Data in the context of smart cities. (CO5)
- (b) A Secure system is needed for the Industrial IoT purpose. What are the important things that need to consider while designing a security protocol for such a system ? (CO5)
- (c) Design an IoT System for Data Aggregation in SmartCities. How can security be managed in such system ? (CO6)